

The exact fractional range for commutators along monomial curves

Candidate full solution to Question 1.15 of arXiv:2304.00621

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Status. Candidate full solution, likely valid; human review is needed. The result determines the half-open interval and the literal supremum asked for in the source. It does not decide boundedness at the non-diagonal boundary itself, which is immaterial to both requested quantities.

1 Source question

Let

$$\gamma(t) = \begin{cases} (\varepsilon_1 |t|^{\beta_1}, \dots, \varepsilon_n |t|^{\beta_n}), & t > 0, \\ (\delta_1 |t|^{\beta_1}, \dots, \delta_n |t|^{\beta_n}), & t < 0, \end{cases} \quad 0 < \beta_1 < \dots < \beta_n,$$

be one of the monomial curves in Oikari's source paper [1], and put

$$H_\gamma f(x) = \text{p.v.} \int_{\mathbb{R}} f(x - \gamma(t)) \frac{dt}{t}.$$

For $p \leq q < \infty$, set

$$B := |\beta| = \sum_{j=1}^n \beta_j, \quad \alpha = B \left(\frac{1}{p} - \frac{1}{q} \right).$$

The estimate under investigation is

$$\| [b, H_\gamma] \|_{L^p(\mathbb{R}^n) \rightarrow L^q(\mathbb{R}^n)} \leq C_{p,q,\gamma} \|b\|_{\text{BMO}^{\beta,\alpha}(\mathbb{R}^n)} \quad \text{for every } b \in \text{BMO}^{\beta,\alpha}. \quad (1)$$

Question 1.15 asks for the maximal initial interval of q for which (1) holds and defines

$$p_{\max}(n, \beta) = \sup\{q \geq p : (1) \text{ holds}\}.$$

1.14. Theorem ([24]). *Let $\gamma : \mathbb{R} \rightarrow \mathbb{R}^2$ be a monomial curve with parts in adjacent quadrants., i.e. $\varepsilon_i = \delta_i$ for exactly one index $i \in \{1, 2\}$. Let $p \leq q$ and $\alpha/|\beta| = 1/p - 1/q$. Then,*

$$\|b\|_{\text{BMO}^{\gamma, \alpha}(\mathbb{R}^2)} \lesssim \| [b, H_\gamma] \|_{L^p(\mathbb{R}^2) \rightarrow L^q(\mathbb{R}^2)}.$$

Already from Theorems 1.10 and 1.14 we see a discrepancy in the obtained upper and lower bounds. The specific form of $\Xi(n)$ stems from the fact that for these exponents we have access to a sparse form domination, which in turn is determined by the availability of smoothing estimates for single scale averaging operators. We do not know if $\Xi(n)$ is maximal with respect to the sufficiency results, but we know from [5, Section 4.2.] that $\Omega(n)$ is sharp with respect to obtaining a sparse form domination of H_γ . By the heuristic principle that commutators of singular integrals are more singular than the singular integrals themselves, we cannot expect a better sparse form domination for the commutator than for the singular integral. However, this does not exclude the possibility that $\Xi(n)$, respectively $p(\Xi(n))$, might not be maximal for the upper bound (1.11).

1.15. Question. *Let $p \in (1, \infty)$, $n \geq 2$ and $\beta \in \mathbb{R}_+^n$ be fixed (i.e. γ is fixed). Determine the maximal interval*

$$(1.16) \quad [p, p_{\max}(n, \beta)), \quad p_{\max}(n, \beta) := \sup\{q \geq p : (1.11) \text{ holds}\}.$$

Does there hold that $p_{\max}(n, \beta) = \infty$?

Now we turn to necessity. One of the two main approaches to commutator lower bounds is through the approximate weak factorization (awf) argument attributable to

Figure 1: Question 1.15 and its surrounding sharpness discussion [1, p. 5].

There is a small logical issue in the question: a supremum over a disconnected validity set need not be the endpoint of a maximal interval. The theorem below determines both notions and shows that they really do diverge for some parameters.

2 The answer

Define

$$p_1(n) := \frac{n(n+1)}{n^2 - n + 2}, \quad p_2(n) := \frac{n+1}{2},$$

and

$$Q_n(p) := \begin{cases} \frac{(n-1)p}{n-p}, & 1 < p \leq p_1(n), \\ \frac{n(n+1)p}{n(n+1)-2p}, & p_1(n) \leq p \leq p_2(n), \\ \frac{np}{n-1}, & p \geq p_2(n). \end{cases} \quad (2)$$

At the shared endpoints the formulas agree. When $n = 2$, the middle interval collapses and

$$Q_2(p) = \begin{cases} p/(2-p), & 1 < p \leq 3/2, \\ 2p, & p \geq 3/2. \end{cases}$$

Theorem 1 (exact range). *For every dimension $n \geq 2$, every monomial exponent tuple β , every allowed sign pattern, and every $1 < p < \infty$, the following hold.*

1. *The estimate (1) holds for $p \leq q < Q_n(p)$.*
2. *If $q > Q_n(p)$ and $\alpha \leq \beta_n$, then (1) fails.*

3. If $\alpha > \beta_n$, then $\text{BMO}^{\beta,\alpha}(\mathbb{R}^n)$ consists only of constants modulo null sets, and (1) holds trivially.

Consequently, the maximal initial half-open interval is exactly

$$[p, Q_n(p)).$$

For the literal supremum displayed in Question 1.15,

$$p_{\max}(n, \beta) = \begin{cases} Q_n(p), & p \geq B/\beta_n, \\ \infty, & 1 < p < B/\beta_n. \end{cases} \quad (3)$$

In the second case, if

$$q_{\text{triv}}(p, \beta) := \frac{pB}{B - p\beta_n}, \quad (4)$$

then (1) fails for $Q_n(p) < q \leq q_{\text{triv}}(p, \beta)$ and holds trivially for every $q > q_{\text{triv}}(p, \beta)$. Thus the literal value ∞ does not mean that $[p, \infty)$ is an interval of validity.

The theorem leaves open only whether (1) holds at $q = Q_n(p)$. The source asks for a half-open interval and a supremum, so this endpoint question does not change the answer.

3 Idea of the proof

The source proves sufficiency in the interior of a polytope $\Omega(n)$. Cladek and Ou proved that the same polytope is sharp for the positive single-scale curve average [2, Sec. 4.2]: outside $\Omega(n)$ there are normalized inputs supported as close to the origin as desired whose averages blow up, with dual witnesses supported near a fixed curve arc.

The key observation is that those separated supports can be encoded into the symbol. Choose $b(x) = \eta(x_n)$, zero on the input neighborhood and one on the output neighborhood. Then $bf = 0$ and $bg = g$, so the commutator pairing is exactly the Hilbert-transform pairing. If $\alpha \leq \beta_n$, this separator is anisotropically α -Hölder and hence belongs to $\text{BMO}^{\beta,\alpha}$. Every sharp single-scale obstruction therefore becomes a commutator obstruction. This transfers all facets of $\Omega(n)$ at once, including the middle facet present when $n > 2$.

There is one genuine twist. If $\alpha > \beta_n$, anisotropic α -Hölder regularity has ordinary exponent greater than one along every coordinate line. Subdividing a coordinate segment forces zero variation, so every symbol is constant. This creates a trivial high- q validity tail and explains the distinction between a maximal initial interval and the literal supremum in the source.

4 The improving polytope and its vertical boundary

Let $\Omega(n)$ be the closed convex hull of

$$(0, 0), \quad (1, 1), \quad A_n = \left(\frac{n^2 - n + 2}{n(n+1)}, \frac{n-1}{n+1} \right), \quad D_n = \left(\frac{2}{n+1}, \frac{2(n-1)}{n(n+1)} \right).$$

These are the coordinates $(1/p, 1/q)$ used in the source. In the half-plane $1/p \geq 1/q$, its lower boundary consists of the three line segments

$$y = \frac{n-1}{n}x, \quad y = x - \frac{2}{n(n+1)}, \quad y = \frac{nx-1}{n-1}. \quad (5)$$

Their x -intervals are respectively

$$\left[0, \frac{2}{n+1}\right], \quad \left[\frac{2}{n+1}, \frac{n^2 - n + 2}{n(n+1)}\right], \quad \left[\frac{n^2 - n + 2}{n(n+1)}, 1\right].$$

Putting $x = 1/p$ and $y = 1/q$ in (5) gives exactly (2). Hence

$$Q_n(p) = \sup\{q \geq p : (1/p, 1/q) \in \Omega(n)\}. \quad (6)$$

Oikari's Theorem 1.10 proves (1) whenever $(1/p, 1/q)$ is in the interior of $\Omega(n)$, as well as on the diagonal $q = p$ [1]. This proves part 1 of Theorem 1.

5 The separating symbol

Use the anisotropic quasi-distance

$$\rho_\beta(x, y) \asymp \max_{1 \leq j \leq n} |x_j - y_j|^{1/\beta_j}.$$

The source's Lemma 3.8 gives

$$\|b\|_{\text{BMO}^{\beta, \alpha}} \asymp \sup_{x \neq y} \frac{|b(x) - b(y)|}{\rho_\beta(x, y)^\alpha} \quad (\alpha > 0). \quad (7)$$

Lemma 2 (separator). *Let $0 < \alpha \leq \beta_n$. There is a smooth bounded function $b(x) = \eta(x_n)$ which is zero on a neighborhood of the origin, one on a neighborhood of the positive curve arc*

$$\Gamma_+ = \{\gamma(t) : 1/2 \leq t \leq 1\},$$

and belongs to $\text{BMO}^{\beta, \alpha}(\mathbb{R}^n)$.

Proof. On the positive branch, the last coordinate of Γ_+ lies in the signed interval with absolute values between $2^{-\beta_n}$ and 1. It is therefore separated from zero. Choose $\eta \in C^\infty(\mathbb{R})$ which is zero on a small interval about zero and one on a small neighborhood of that signed interval.

Since η is bounded and Lipschitz,

$$|\eta(x_n) - \eta(y_n)| \lesssim \min(1, |x_n - y_n|).$$

If $\rho_\beta(x, y) \leq 1$, then $|x_n - y_n| \lesssim \rho_\beta(x, y)^{\beta_n} \leq \rho_\beta(x, y)^\alpha$. If $\rho_\beta(x, y) \geq 1$, boundedness of η gives the same conclusion. Now use (7). $\square$

6 Transfer of the sharp single-scale obstruction

Lemma 3 (separated-support transfer). *If $(1/p, 1/q)$ lies strictly outside $\Omega(n)$ and $0 < \alpha \leq \beta_n$, then (1) fails.*

Proof. For the positive single-scale operator

$$\tilde{A}_1^\gamma f(x) = \int_{1/2 \leq t < 1} f(x - \gamma(t)) dt,$$

Cladek and Ou prove the following sharpness statement [2, Sec. 4.2]. For every sufficiently small neighborhood U of the origin there are nonnegative f_k supported in U with $\|f_k\|_p = 1$ and

$$\|\tilde{A}_1^\gamma f_k\|_q \longrightarrow \infty.$$

By duality there are nonnegative g_k , with $\|g_k\|_{q'} = 1$, supported in an arbitrarily small neighborhood V of Γ_+ , such that the corresponding Hilbert-transform pairings tend to infinity. Their support argument isolates the positive branch and the finitely many relevant dyadic scales, so in particular

$$\langle H_\gamma f_k, g_k \rangle \longrightarrow \infty. \quad (8)$$

Choose U, V small enough for the symbol in Lemma 2 to satisfy $b = 0$ on U and $b = 1$ on V . Then $bf_k = 0$ and $bg_k = g_k$. Consequently,

$$\begin{aligned} \langle [b, H_\gamma] f_k, g_k \rangle &= \langle b H_\gamma f_k, g_k \rangle - \langle H_\gamma (b f_k), g_k \rangle \\ &= \langle H_\gamma f_k, g_k \rangle \longrightarrow \infty. \end{aligned}$$

If (1) held, Hölder duality and the normalizations would instead give

$$|\langle [b, H_\gamma] f_k, g_k \rangle| \leq C_{p,q,\gamma} \|b\|_{\text{BMO}^{\beta,\alpha}} \quad \text{uniformly in } k,$$

a contradiction. $\square$

For $q > Q_n(p)$, equation (6) places $(1/p, 1/q)$ strictly outside $\Omega(n)$. Lemma 3 proves part 2 of Theorem 1.

7 Collapse above the largest anisotropic exponent

Lemma 4 (triviality). *If $\alpha > \beta_n$, every element of $\text{BMO}^{\beta,\alpha}(\mathbb{R}^n)$ is constant modulo null sets.*

Proof. By (7), choose the anisotropic Hölder representative of b . Along the j th coordinate line,

$$|b(x + he_j) - b(x)| \lesssim |h|^{\alpha/\beta_j}.$$

Here $\alpha/\beta_j > 1$ for every j . Divide the segment from x to $x + he_j$ into N equal pieces. The triangle inequality gives

$$|b(x + he_j) - b(x)| \lesssim N \left(\frac{|h|}{N} \right)^{\alpha/\beta_j} = |h|^{\alpha/\beta_j} N^{1-\alpha/\beta_j} \longrightarrow 0.$$

Thus b is constant on every coordinate line, hence on $\mathbb{R}^n$. $\square$

Constants commute with H_γ , proving part 3. It remains only to extract (3). If $p \geq B/\beta_n$, then for every finite q ,

$$\alpha = B(1/p - 1/q) < B/p \leq \beta_n.$$

Part 2 therefore excludes every $q > Q_n(p)$, and the supremum is $Q_n(p)$.

If $p < B/\beta_n$, equation $\alpha = \beta_n$ has the finite solution (4). For $q > q_{\text{triv}}$, Lemma 4 makes (1) trivial, so the literal supremum is infinity. Moreover, $q_{\text{triv}} > Q_n(p)$. Indeed, throughout $\Omega(n)$ one has

$$0 \leq x - y \leq \frac{2}{n(n+1)},$$

and, because the exponents are strictly increasing,

$$B \frac{2}{n(n+1)} < \frac{2\beta_n}{n+1} < \beta_n.$$

Thus $\alpha < \beta_n$ at the vertical lower boundary, while α grows strictly with q and reaches β_n only at q_{triv} . Part 2 now gives failure on $Q_n(p) < q \leq q_{\text{triv}}$, completing the proof of Theorem 1.

8 Verification, upgrade audit, and novelty bounds

The proof was developed through six focused routes. A direct ball/tube test first recovered both facets for the parabola. The adjoint test and a one-coordinate separator extended this to every planar monomial curve. In higher dimensions the same test recovered only the two outer facets and missed the middle one. The successful upgrade was to transfer the complete sharp single-scale sequence from Cladek–Ou, which supplies every facet in every dimension. A separate audit of high Hölder orders found the collapse of the symbol space and the resulting defect in the literal supremum. The remaining boundary case was checked and is not needed for the half-open interval or its supremum.

The run indexes were searched for arXiv:2304.00621, “monomial curve commutator”, “maximal interval”, “fractional BMO”, and the exact wording of Question 1.15. Bounded primary-source web and arXiv searches through 13 August 2026 checked the source, arXiv:1704.07810, and later curved commutator work including arXiv:2403.08338. The latter completes the planar $p = q$ lower-bound theory, not the fractional sufficiency range. No paper was found that states the theorem above or answers Question 1.15. Novelty confidence is moderate because the literature search was bounded.

Promote for expert review. The most important review point is that the precise support localization in Cladek–Ou Section 4.2 is strong enough to choose one separator with $bf_k = 0$ and $bg_k = g_k$ throughout the sequence. The second point is the distinction between the initial interval endpoint and the unrestricted supremum when the Campanato class becomes constant.

References

- [1] T. Oikari, *On the L^p -to- L^q boundedness and compactness of commutators along monomial curves*, arXiv:2304.00621 (2023).
- [2] L. Cladek and Y. Ou, *Sparse domination of Hilbert transforms along curves*, Math. Res. Lett. 25 (2018), no. 2, 415–436; arXiv:1704.07810.